Water

1. When water dissolves a substance like salt, which atom of the water molecule attaches to the positive sodium cation? Why?

2. How many hydrogen bonds can each oxygen atom make?

3. How many hydrogen bonds can each hydrogen atom make?

Macromolecules and Functional Groups

4. What element forms the skeleton of all organic macromolecules? Why?

5. What additional elements are found in functional groups that are attached to the skeleton structure?

6. How many carbon atoms are in each depicted molecule below.

a) 

b) 

c) 

7. What kind of chemical reaction links together two monomers? Is water a product or reactant in this kind of reaction?

8. What kind of chemical reaction breaks apart a macromolecule? Is water a product or reactant in this kind of reaction?

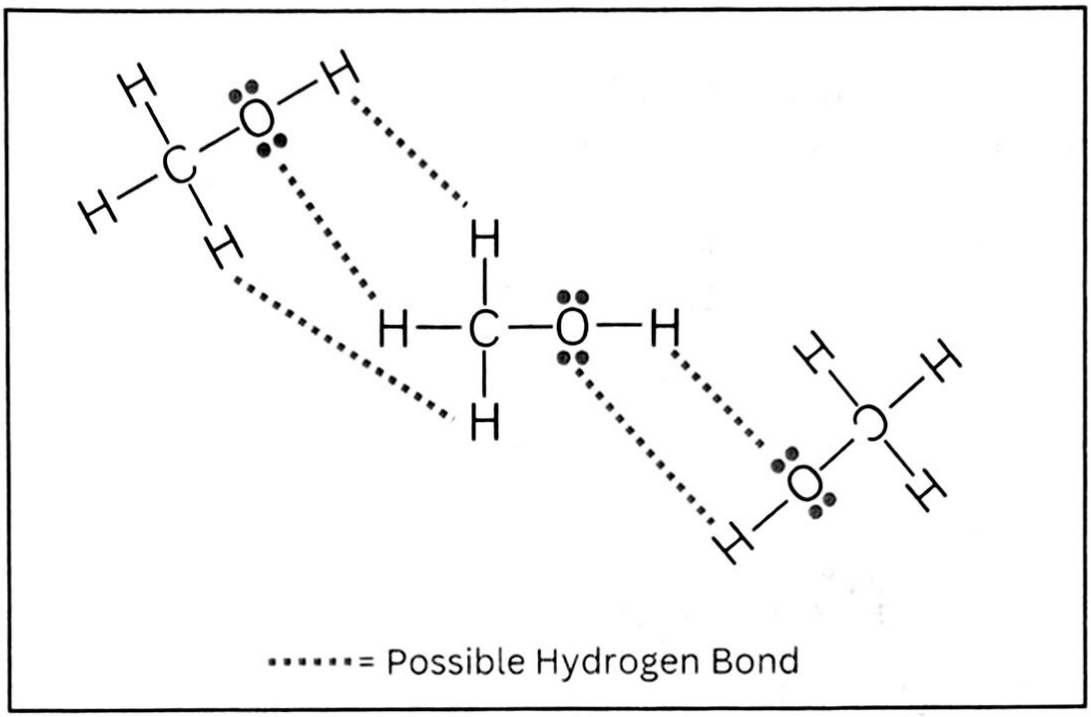
9. How many bonds does oxygen form in its most stable configuration? Nitrogen?

10. Explain the difference between the bonds in CO_2 and the bonds in NaCl

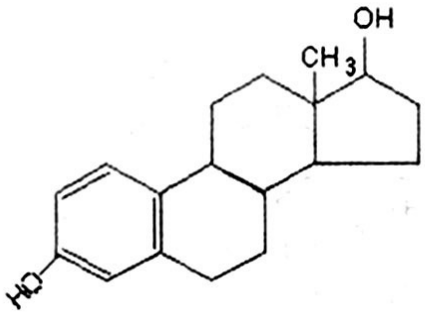
11. Place a Y in the chart into each space below if biological molecules of that category contain atoms of that element

	Carbon	Hydrogen	Oxygen	Nitrogen	Phosphorous	Sulfur
Carbohydrates						
Lipids (triglyceride)						
Protein						
Nucleic acid						
Glucose						
Enzymes						
DNA						
Phospholipids						
Steroids						

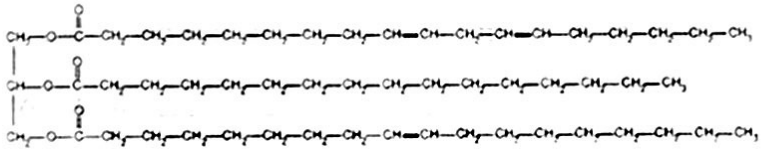
12. Circle the *correctly* drawn hydrogen bonds in the diagram below



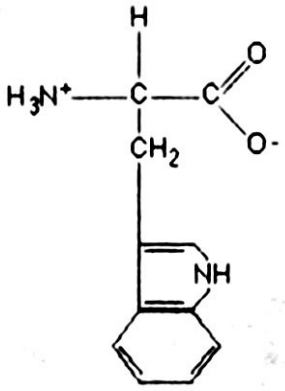
13. Identify which category the following macromolecules are from



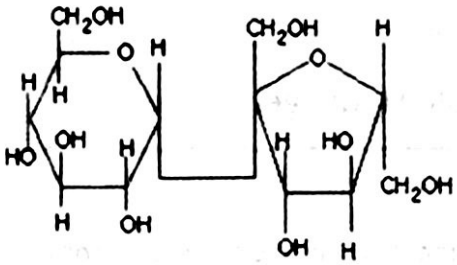
a) _____



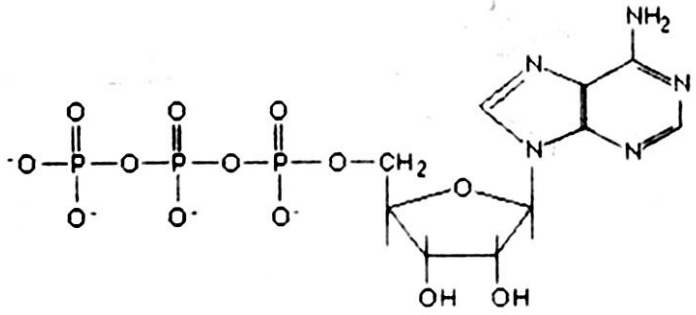
b) _____



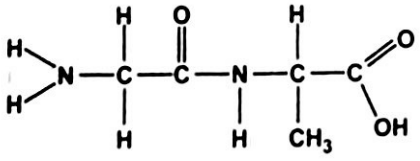
c) _____



d) _____



e) _____



f) _____

14. For each row, put a (Y) in the proper column to identify the functional group described. Some terms may apply to more than one group

	Alcohol (hydroxyl)	Amine (Amino)	Carboxylic Acid	Phosphate
-OH				
$-NH_2$				
-COOH				
$-PO_4^{-2}$				
Polar				
Acidic				
Basic				
Found in carbohydrates				
Found in <u>all</u> amino acids				
Found in <u>some</u> amino acids				
Found in phospholipids				
Found in nucleic acids				

15. Identify the monomer for each class of macromolecules

Carbohydrates: _____

Nucleic acids: _____

Proteins: _____

16. Given the function/description, place a (Y) in the box under the macromolecule it matches

	Carbohydrate	Lipid	Protein	Nucleic Acid
Has both hydrophobic and hydrophilic properties				
Contains the genetic code				
Steroids				
Used for short term energy				
Makes up plant cell walls				
Long term energy storage				
Make up the cell membrane				

Proteins

Indicate the level of protein structure that each characteristic is describing (Choices – primary, secondary, tertiary, quaternary)

17. Can be either an alpha helix or beta pleated sheet

18. Bonds at this level are called peptide bonds

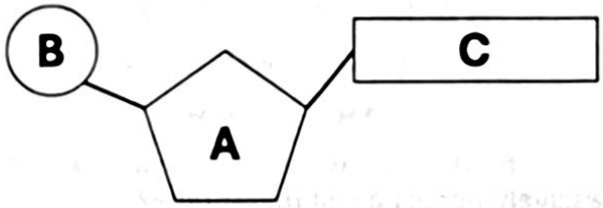
19. Bonds at this level are hydrophobic interactions, disulfide bridges, and intermolecular forces

20. This is the level of folding that is made of several polypeptide chains linked together (ex. hemoglobin – four chains lined together)

21. DNA directly codes for this level of structure

Nucleic Acids

22. Each monomer of a nucleic acid has 3 parts, identify them below



A) _____

B) _____

C) _____

23. The nitrogenous bases in DNA molecules are held together by _____

24. The backbone of DNA and RNA molecules are held together by _____

25. Why are DNA sequences often listed with bas pairs in groups of 3?
(Ex: 3' ATA TAT GCT ACT GCA AAG 5')

Carbohydrates

26. Structural polysaccharides have a _____ structure.

In this structure the bulky CH_2OH groups extend on _____ sides of the rings.

27. Energy storage polysaccharides have a _____ structure.

In this structure the bulky CH_2OH groups extend on _____ sides of the rings.

28. In carbohydrate molecules, the elements C, H, and O are present in a _____ ratio.

Lipids

29. Steroids have a structure comprised of _____ and a side chain, and serve primarily as _____

30. What type of molecules can diffused passively across a cell membrane?

31. Which region of phospholipids are oriented towards the cellular environment? Why?

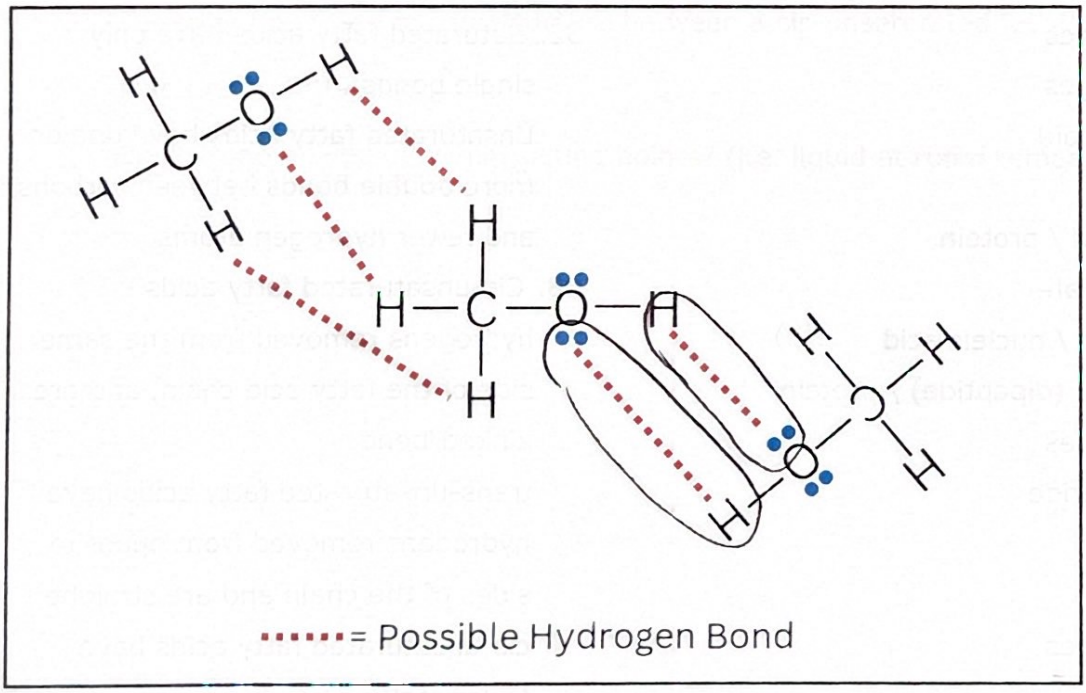
32. Describe the difference between a saturated and unsaturated fatty acid

33. Describe the difference in structure between a cis-unsaturated fatty acid, and a trans-unsaturated fatty acid.

Which generally have lower melting points? (i.e. liquid at room temperature)

1. Oxygen atoms, since they have a partial negative charge
2. 2
3. 1
4. Carbon, because it can form four bonds – it creates long continuous chains with diverse shapes
5. Oxygen, nitrogen, phosphorous (and sulfur)
6. a) 6 b) 4 c) 5
7. Dehydration synthesis, water is a product
8. Hydrolysis, water is a reactant
9. Oxygen – 2 bonds
   Nitrogen – 3 bonds
10. CO_2 has covalent bonds where atoms share electrons, NaCl has ionic bonds where electrons have moved from one atom to another, creating ions
11. See next pages
12. See next pages
13. a) lipid, steroid
    b) lipid, fat
    c) amino acid / protein
    d) carbohydrate
    e) nucleotide / nucleic acid
    f) amino acid (dipeptide) / protein
14. See next pages
15. Monosaccharide
    nucleotide
    amino acid
16. See next pages
17. Secondary
18. Primary
19. Tertiary
20. Quaternary
21. Primary
22. a) 5-carbon sugar
    b) Phosphate group
    c) Nitrogenous base
23. Hydrogen bonds
24. Covalent bonds
25. DNA is read in codons of 3 base pairs at a time
26. Linear, alternating
27. Helical, same
28. 1:2:1
29. Four fused carbon rings, signaling molecules
30. Small non-polar molecules
31. The polar head group (phosphate), favorably interacts with the water-based intracellular environment
32. Saturated fatty acids have only single bonds
    Unsaturated fatty acids have one or more double bonds between carbons, and fewer hydrogen atoms
33. Cis-unsaturated fatty acids - hydrogens removed from the same side of the fatty acid chain, and are kinked/bent
    trans-unsaturated fatty acids have hydrogens removed from opposite sides of the chain and are straight
    cis-unsaturated fatty acids have lower melting points

	Carbon	Hydrogen	Oxygen	Nitrogen	Phosphorous	Sulfur
Carbohydrates	X	X	X			
Lipids (triglyceride)	X	X	X			
Protein	X	X	X	X		X
Nucleic acid	X	X	X	X	X	
Glucose	X	X	X			
Enzymes	X	X	X	X		X
DNA	X	X	X	X	X	
Phospholipids	X	X	X		X	
Steroids	X	X	X			



Emet Classical Academy

Answer Key

10th Grade Integrated Sciene

Unit 1 – The Chemistry of Life, Review

	Alcohol (hydroxyl)	Amine (Amino)	Carboxylic Acid	Phosphate
-OH	X			
$-NH_2$		X		
-COOH			X	
$-PO_4^{-2}$				X
Polar	X	X	X	X
Acidic			X	
Basic		X		
Found in carbohydrates	X			
Found in all amino acids		X	X	
Found in some amino acids	X			
Found in phospholipids				X
Found in nucleic acids	X	X (rings)		X

	Carbohydrate	Lipid	Protein	Nucleic Acid
Has both hydrophobic and hydrophilic properties		X (phospholipid)	X (some)	
Contains the genetic code				X
Steroids		X		
Used for short term energy	X			
Makes up plant cell walls	X			
Long term energy storage		X (fats)		
Make up the cell membrane		X		